INFORMATION COMMUNICATION TECHNOLOGY

SOFTWARE PRACTICAL 3

TERM 3,MODULE 3

UNIT CODE: 0619 451 06A

ELEMENT 2: TEST COMPUTER SOFTWARE FUNCTIONALITY

Time: 2 hours

Free API to use: https://jsonplaceholder.typicode.com

Instructions

1. *Open Postman and create a **new Collection** named YourName_REST_Practical.*
2. *Save **every** request inside this collection, and name each request clearly (e.g., Task 1 - GET All Users).*
3. *For each task, take a **screenshot** showing the method, URL, request body (where applicable), status code, and response.*
4. *Write your answers to the questions in a document and submit it together with your exported collection (JSON) and screenshots.*
5. *Note: JSONPlaceholder simulates changes. Created, updated, or deleted data is not actually saved, so a GET afterwards will not show your changes. This is normal.*

Section A: GET Requests

Task 1: Retrieve all users

- Send a GET request to https://jsonplaceholder.typicode.com/users
- Screenshot the response showing the status code, response time, and size.
- Question: How many users were returned?

Task 2: Retrieve a single user

- Send a GET request to https://jsonplaceholder.typicode.com/users/3
- Write down the name, username, and email of this user.

Task 3: Query parameters

- Send a GET request to https://jsonplaceholder.typicode.com/posts
- Using the **Params** tab (not by typing in the URL), add the key userId with the value 2.
- Record the number of posts returned.

Section B: POST Request

Task 4: Create a new post

- Method: POST
- URL: https://jsonplaceholder.typicode.com/posts
- Under **Body → raw → JSON**, enter:

```
{
  "title": "My Practical Post",
```

```
 "body": "Created during the Postman practical test",
 "userId": 5
}
```

- Click **Send**.
- Questions:
    1. What status code was returned, and what does it mean?
    2. What id did the server assign to the new record?
- Include a screenshot of both the request and the response.

Section C: PUT and PATCH Requests

Task 5: Update a post using PUT

- Method: PUT
- URL: https://jsonplaceholder.typicode.com/posts/1
- Body (JSON):

```
{
 "id": 1,
 "title": "Updated Title by [Your Name]",
 "body": "This post has been fully updated",
 "userId": 1
}
```

- Confirm that the response shows the updated values and record the status code.

Task 6: Partially update using PATCH

- Method: PATCH
- URL: https://jsonplaceholder.typicode.com/posts/1
- Body (JSON): change **only** the title:

```
{
```

```
 "title": "Patched Title"
}
```

- Question: In one or two sentences, explain the difference between PUT and PATCH.

Section D: DELETE Request

Task 7: Delete a post

- Method: DELETE
- URL: https://jsonplaceholder.typicode.com/posts/1
- Record the status code and explain what it means.

Section E: Status Codes and Error Handling

Task 8: Trigger a 404 error

- Send a GET request to https://jsonplaceholder.typicode.com/users/9999
- Record the status code and the response body.
- Explain why this error occurred.

Task 9: Match the status codes

Write the meaning of each code in your own words:

Code	**Meaning**
200	
201	
400	

401	
403	
500	

Section F: Organisation and Automation

Task 10: Variables and test scripts

1. Create a **Collection variable** named baseUrl with the value https://jsonplaceholder.typicode.com.
2. Edit your Task 1 request to use {{baseUrl}}/users as the URL.
3. In the **Tests** tab, add this script and resend the request:

```
pm.test("Status code is 200", function () {
   pm.response.to.have.status(200);
});
```

4. Take a screenshot showing the test as **Passed** in the Test Results tab.